

USP General Chapter <800> Hazardous Drugs — Handling in Healthcare Settings

A Training Module for the Compounding Academy

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Learning Objectives

By the end of this module, you will be able to:

1

Define & Identify

Define a hazardous drug (HD) using the NIOSH criteria and identify the scope of <800>.

2

Apply AoR

Apply the entity's HD list and the Assessment of Risk (AoR) framework to appropriate dosage forms.

3

Differentiate Controls

Differentiate the three levels of engineering controls (C-PEC, C-SEC, supplemental) and the two acceptable sterile HD configurations.

4

Select PPE

Select appropriate PPE for each HD activity, including ASTM D6978 chemotherapy gloves.

5

Execute Procedures

Execute receipt, compounding, administration, deactivation, spill, and medical-surveillance procedures in compliance with <800>.

 <800> is an occupational safety standard — its first goal is protecting the worker.

Scope of <800>

Three Protected Populations

- Patient safety
- Worker safety
- Environmental protection

Activities Covered

Receipt, storage, compounding, dispensing, administration, and disposal of sterile and nonsterile HD products and preparations.

Who Must Comply

All healthcare personnel and all entities that store, prepare, transport, or administer HDs — pharmacies, hospitals, clinics, physicians' and veterinarians' offices.

Mandatory Occupational Safety Plan Elements

HD List

Facility & Engineering
Controls

Competent Personnel

Safe Work Practices

Proper PPE Use

HD Waste Segregation & Disposal

⚠️ <800> is a system, not a checklist — every entity must build all six program elements.

The NIOSH List & Containment Requirements

An entity must maintain its own HD list, drawn from the current NIOSH List, and review it at least every 12 months.

MUST Follow ALL Containment Requirements

- Any HD active pharmaceutical ingredient (API)
- Any antineoplastic requiring HD manipulation

MAY Use Assessment of Risk for Alternative Containment

- Final dosage forms of compounded HD preparations and conventionally manufactured HD products — including antineoplastic dosage forms requiring no manipulation beyond counting or repackaging (unless required by manufacturer)
- Other dosage forms of HDs on the NIOSH list

⊗ If insufficient information exists about a drug, treat it as hazardous until proven otherwise.

⚠ APIs and antineoplastics needing manipulation always require full containment — no exceptions.

Assessment of Risk

An AoR may justify alternative containment for eligible HDs — but it must be documented and reviewed **annually**.

At Minimum, an AoR Must Consider

- Type of HD (antineoplastic, non-antineoplastic, reproductive risk only)
- Dosage form
- Risk of exposure
- Packaging
- Manipulation required

Documentation Requirements

- What alternative containment strategies and/or work practices are being used
- Which specific dosage forms each strategy applies to
- Annual review with documented outcome

 If you don't perform an AoR, you must apply ALL <800> containment strategies to every HD.

 The AoR is your only path to right-sized containment for low-risk HD dosage forms.

How HDs Enter the Body

HDs reach personnel through unintentional routes — contamination is often invisible.

1

Dermal Absorption

Skin contact with residue on vials, surfaces, and PPE.

2

Mucosal Absorption

Contact with eyes, lips, and nasal membranes.

3

Inhalation

Aerosols, vapors, and powder dust generated during handling.

4

Injection

Needlestick injuries during compounding or administration.

5

Ingestion

Contaminated foodstuffs or hand-to-mouth contact.

⊗ Containers of HDs have been shown to be contaminated upon receipt. Both clinical AND nonclinical personnel may be exposed when handling HDs or contaminated surfaces.

⚠ Assume every HD container, surface, and piece of PPE is contaminated until proven otherwise.

Activities With Exposure Potential

Table 1 — Examples of Potential Opportunities of Exposure

Activity	Potential Opportunity of Exposure
Receipt	HD residue on containers, dosage units, outer cartons, work surfaces, floors
Dispensing	Counting or repackaging tablets and capsules
Compounding & Manipulation	Crushing/splitting tablets, opening capsules; pouring liquids; weighing/mixing; reconstituting powders; withdrawing or diluting injectables; expelling air from syringes; residue on PPE
Cleaning & Maintenance	Deactivating, decontaminating, cleaning areas contaminated with HDs; maintaining contaminated equipment
Administration	Aerosols from injection, irrigation, oral, inhalation, topical routes; specialized procedures (intraperitoneal, bladder instillation); IV priming
Patient Care	Body fluids (urine, feces, sweat, vomit) and contaminated linens, dressings, clothing
Spills	Spill generation, management, and disposal
Transport	Moving HDs within the healthcare setting
Waste	Collecting and disposing hazardous and trace-contaminated waste

 Exposure isn't limited to compounding — receipt, administration, and waste handling are equally high-risk.

The Designated Person

Every entity must name a qualified, trained designated person accountable for the HD program.

The Designated Person Must:

- Develop and implement HD procedures
- Oversee entity compliance with <800> and applicable laws and standards
- Ensure competency of all personnel handling HDs
- Ensure environmental control of HD storage and compounding areas
- Monitor the facility and maintain reports of testing and sampling; act on the results
- Understand the rationale for risk-prevention policies and consequences of non-compliance
- Report potentially hazardous situations to management

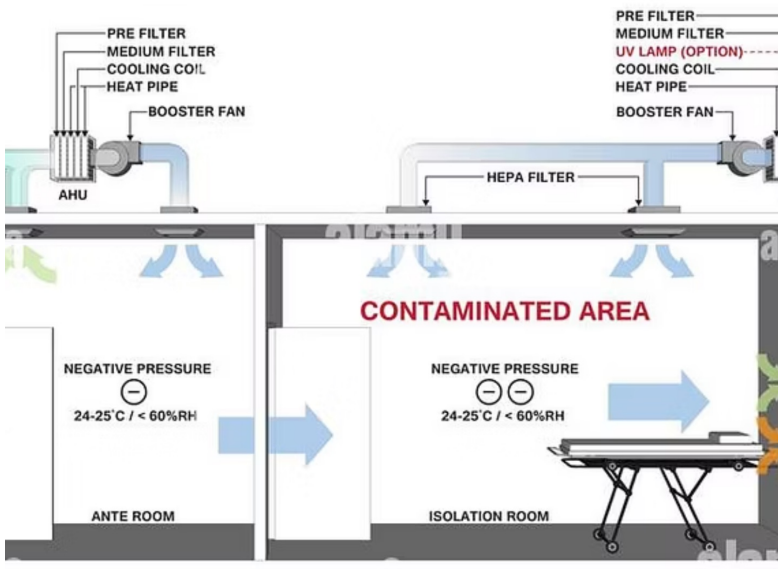
All Personnel Handling HDs Share Responsibility For:

- Understanding fundamental practices and precautions
- Continually evaluating procedures and final HD quality
- Preventing patient harm, minimizing personnel exposure, and minimizing environmental contamination



Accountability is named and documented — and shared by everyone who touches an HD.

NEGATIVE ROOM PRESSURE



Facilities — Designated Areas

General Requirements

- Hazard signage prominently posted before entry to HD handling areas
- Access restricted to authorized personnel
- HD handling areas located AWAY from breakrooms and refreshment areas
- Consider uninterrupted power supplies (UPS) to preserve negative pressure during outages

Required Designated Areas

- Receipt and unpacking
- Storage of HDs
- Nonsterile HD compounding (if performed)
- Sterile HD compounding (if performed)

⊗ Certain areas **MUST** be **negative pressure** relative to surrounding areas.

⚠ Containment starts with how the building itself is laid out.

Receipt and Storage

Receipt Area (§5.1)

- Antineoplastic HDs and all HD APIs must be unpacked in an area that is neutral/normal or **NEGATIVE** pressure relative to surroundings
- HDs must **NOT** be unpacked in sterile compounding areas or positive-pressure areas

Storage Area (§5.2)

- Store to prevent spillage or breakage if a container falls
- Do **NOT** store HDs on the floor
- In disaster-prone regions: secure shelves with raised front lips
- Antineoplastic HDs requiring manipulation AND all HD APIs must be stored separately in an externally vented, negative-pressure room with ≥ 12 ACPH
- Non-antineoplastic, reproductive-risk-only, and final dosage forms of antineoplastics MAY be stored with other inventory if entity policy permits
- Refrigerated antineoplastic HDs: dedicated refrigerator in a negative-pressure area with ≥ 12 ACPH; consider an exhaust adjacent to the compressor

⊗ **12 ACPH** is the minimum for storage of antineoplastics requiring manipulation.

⚠ Pressure direction and dedicated storage are non-negotiable for antineoplastics and HD APIs.

Engineering Controls — Three Levels

Level 1 — Primary (C-PEC)

A ventilated device that minimizes worker and environmental HD exposure during direct handling.

Examples: CVE, Class I/II BSC, CACI

Level 2 — Secondary (C-SEC)

The room in which the C-PEC is placed. Must be externally vented, physically separated, properly ventilated, and at negative pressure **0.01–0.03 in. w.c.** relative to all adjacent areas.

Level 3 — Supplemental

Adjunct controls offering additional protection.

Example: Closed-system drug-transfer device (CSTD)

⊗ A CSTD is **NEVER** a substitute for a C-PEC.

⚠ C-PEC inside C-SEC, with CSTDs layered on top — three lines of defense.

Nonsterile HD Compounding

🔗 ALSO SEE USP <795>

Must additionally follow <795>. Acceptable C-PECs include: Class I BSC or CVE (preferred for nonsterile), Class II BSC or CACI.

📌 A C-PEC is **NOT required** if manipulation is limited to final dosage forms (counting, repackaging) that do not produce particles, aerosols, or gases.

Table 2 — Nonsterile HD Compounding: C-PEC and C-SEC Requirements

C-PEC	C-SEC Requirements
Externally vented (preferred) OR redundant HEPA filters in series. Examples: CVE, Class I or II BSC, CACI	Externally vented · ≥ 12 ACPH · Negative pressure 0.01–0.03 in. w.c. relative to adjacent areas

Surface Requirements

Ceilings, walls, floors, fixtures, shelving, counters, cabinets: smooth, impervious, free from cracks and crevices, non-shedding.

⚠ For nonsterile HD work the C-PEC doesn't need ISO classification — but the room still needs 12 ACPH and negative pressure.

Sterile HD — ISO 7 Buffer + ISO 7 Ante (Preferred)

ALSO SEE USP <797>

Must additionally follow <797>. All C-PECs for sterile HDs must be externally vented and provide ISO Class 5 or better.

Acceptable C-PECs

Class II BSC types A2, B1, or B2 (A2 most common) · Class III BSC · CACI

⊗ DO NOT use LAFW or CAI for compounding antineoplastic HDs. A BSC/CACI used for HDs must not prepare a non-HD (unless wrapped in protective outer wrapper with PPE-handling label).

Table 3 (Row 1) — ISO 7 Buffer Room Configuration

C-PEC	C-SEC	Maximum BUD
Externally vented (Class II BSC or CACI)	Externally vented · 30 ACPH · Negative pressure 0.01–0.03 in. w.c. relative to adjacent areas · ISO Class 7 buffer room with fixed walls and HEPA-filtered supply	As described in <797>

Ante-Room Requirements

- ≥ **30 ACPH** HEPA-filtered supply
- Positive pressure ≥ **0.02 in. w.c.** relative to adjacent unclassified areas
- ISO Class 7 air quality with fixed walls
- Hand-washing sink ≥ **1 meter** from the entrance to the HD buffer room

⚠ This is the only configuration that lets you use standard <797> BUDs for sterile HD CSPs.

Sterile HD — Containment Segregated Compounding Area

🔍 ALSO SEE USP <797>

An alternative when a full ISO 7 suite is not feasible. Must additionally follow <797>.

Table 3 (Row 2) — C-SCA Configuration

C-PEC	C-SEC	Maximum BUD
Externally vented (Class II BSC or CACI)	Externally vented · ≥ 12 ACPH · Negative pressure 0.01–0.03 in. w.c. relative to adjacent areas · Fixed walls · Unclassified	Limited to BUDs in <797> for CSPs in a segregated compounding area

C-SCA Rules

- Hand-washing sink ≥ 1 meter from the C-PEC; may be inside or directly outside the C-SCA
- Only low- and medium-risk HD CSPs may be prepared in a C-SCA
- All BUDs are shorter than in the ISO 7 configuration

⊗ Choosing the C-SCA buys you simplicity at the cost of BUD.

⚠ C-SCA is a legitimate path — but expect shorter BUDs and a narrower CSP risk-level range.

Closed-System Drug-Transfer Devices (CSTDs)

What a CSTD Does

- Mechanically prohibits transfer of environmental contaminants INTO the system and escape of HD or vapor OUT of the system
- Reduces aerosols during compounding when the dosage form allows

When CSTDs Are Required vs. Recommended

- **MUST** be used when **ADMINISTERING** antineoplastic HDs when the dosage form allows
- **SHOULD** be used when **COMPOUNDING** HDs when the dosage form allows
- Must **NOT** be used if known to be physically or chemically incompatible with the specific HD

⊗ A CSTD is a **supplemental control** — it does **NOT** replace the C-PEC.

⚠ CSTDs add a layer of protection — required at the bedside for antineoplastics.

Environmental Wipe Sampling

Frequency

- Initially as a benchmark
- At least every **6 months** thereafter, or more often as needed to verify containment

Where to Sample

- Interior of the C-PEC and equipment within it
- Pass-through chambers
- Staging and work areas near the C-PEC
- Floors directly under the C-PEC and adjacent dispensing areas
- Immediately outside the HD buffer room or C-SCA
- Patient administration areas

Common Marker HDs

Cyclophosphamide, ifosfamide, methotrexate, fluorouracil, platinum-containing drugs

- ⊗ There is currently no universal acceptance limit, but some studies show worker uptake at cyclophosphamide > **1.00 ng/cm²**.

If Measurable Contamination Is Found

- Designated person must identify, document, and contain the cause
- Actions may include re-evaluating work practices, retraining, thorough deactivation/decontamination/cleaning, and improving engineering controls
- Repeat wipe sampling to validate effectiveness

 Wipe sampling is the only way to prove your containment is actually working.

PPE — Overview and Gloves

Appropriate PPE is required for: receipt, storage, transport, compounding, administration, deactivation/decontamination/cleaning/disinfection, spill control, and waste disposal.

Baseline PPE for Compounding Sterile & Nonsterile HDs

- Gown
- Head, hair, shoe covers
- **TWO pairs of chemotherapy gloves**

Chemotherapy Glove Requirements

- MUST meet **ASTM D6978** (or successor)
- MUST be powder-free
- Inspect for pinholes or weak spots before use — discard if defective
- Outer glove must be **STERILE** when used for sterile compounding
- Change every **30 minutes** (unless manufacturer specifies otherwise) and immediately when torn, punctured, or contaminated
- Wash hands with soap and water after removing gloves

⚠ **ASTM D6978**, powder-free, doubled, and changed every **30 minutes** — that is the chemotherapy glove standard.



PPE — Gowns

Gown Requirements

- Disposable and demonstrated to resist HD permeability
- Polyethylene-coated polypropylene or other laminate materials offer better protection than uncoated
- Close in the back (no open front), long sleeves, closed elastic or knit cuffs, no seams or closures that allow HD passage

NOT Acceptable

- Cloth laboratory coats, surgical scrubs, isolation gowns, or other absorbent materials — they permit HD permeation and hold spills against the skin

Change Frequency

- Per manufacturer's permeation data
- If no permeation data: every **2–3 hours** OR immediately after a spill or splash

Required Activities

- Required for compounding sterile and nonsterile HDs
- Required (permeation-tested) gowns for **ADMINISTERING** injectable antineoplastic HDs

⊗ HD-handling gowns and shoe covers must **NEVER** be worn into other areas — and never taken home.

⚠ A cloth lab coat is not protection — only a permeation-tested disposable gown is.

PPE — Head, Hair, Shoe, Eye, and Face

Head, Hair, Shoe, and Sleeve Covers

- Head and hair covers include beard and mustache covers when applicable
- A **SECOND pair of shoe covers** must be donned before entering the C-SEC and doffed when exiting
- Shoe covers worn in HD areas must not be worn elsewhere
- Disposable sleeve covers (polyethylene-coated polypropylene or laminate) provide added arm protection

Eye and Face Protection

Required when handling HDs outside a C-PEC with risk of spill or splash:

- **Goggles** for eye protection — eyeglasses or safety glasses with side shields do NOT protect adequately
- **Face shield + goggles** = full splash protection
- Face shield alone is **NOT** sufficient eye protection
- A full-facepiece respirator provides combined eye and face protection

 Splash protection requires goggles PLUS a face shield — neither one alone is enough.

PPE — Respiratory Protection

Specific Scenario Use

- **Unpacking HDs not in plastic:** elastomeric half-mask with multi-gas cartridge and P100 filter until packaging integrity verified
- **Most HD activities requiring respiratory protection:** fit-tested NIOSH-certified N95 or more protective

Full-Facepiece or PAPR Required When

- Attending HD spills larger than the spill kit can contain
- Deactivating, decontaminating, cleaning UNDER the work surface of a C-PEC
- Known or suspected airborne exposure to powders or vapors

What Does NOT Work

- Surgical masks do **NOT** provide respiratory protection from HDs
- N95 respirators offer no protection against gases or vapors
- A surgical N95 adds splash protection but is still respiratory-equivalent to a standard N95

❌ Compliance: Fit-test the respirator and train workers. Follow OSHA 29 CFR 1910.134.

⚠️ A surgical mask is not respiratory protection — fit-tested N95 is the floor, PAPR is the ceiling.

Hazard Communication Program

Required under OSHA's Hazard Communication Standard (29 CFR 1910.1200) and aligned with GHS.

→ Written Plan

A written plan describing how the standard is implemented at your entity.

→ Container Labeling

All containers of hazardous chemicals labeled, tagged, or marked with identity and hazard warnings at all times.

→ Safety Data Sheets

An SDS for every hazardous chemical used. SDSs must be readily accessible to personnel during **EVERY work shift** and in their work areas.

→ Training Requirements

Information and training provided **BEFORE** initial assignment and again whenever the hazard changes.

→ Reproductive Risk Acknowledgment

Personnel of reproductive capability must **CONFIRM IN WRITING** that they understand the risks of handling HDs.

⚠ Reproductive-capability acknowledgment must be written — a verbal disclosure is not enough.

Personnel Training

Required Cadence

- Training **before** independent HD handling
- Competency reassessment at least every **12 months**
- Retraining before introducing a new HD, new equipment, or a significant process/SOP change
- All training and competency assessments **documented**

Minimum Training Content

- Overview of the entity's HD list and risks
- Review of HD-related SOPs
- Proper use of PPE
- Proper use of equipment and engineering controls
- Response to known or suspected HD exposure
- Spill management
- Proper disposal of HDs and trace-contaminated materials

 Training is event-driven AND time-driven — annually plus on every meaningful change.

Receiving HDs and Damaged Containers

Receipt Practice

- Receive HDs from the supplier in impervious plastic when possible
- Deliver HDs to the HD storage area **IMMEDIATELY** after unpacking
- Wear PPE including chemotherapy gloves when unpacking
- A spill kit must be accessible in the receiving area
- Visually inspect every shipping container for damage or breakage before opening

Table 4 — Receiving HDs: Damaged Container Procedures

Scenario	Required Action
Damaged shipping container (unopened)	Seal without opening; contact the supplier. If returned, enclose in an impervious container labeled "Hazardous." If supplier declines return, dispose as hazardous waste.
Damaged container that must be opened	Seal in plastic or impervious container; transport to a C-PEC; place on a plastic-backed prep mat; open and remove undamaged items; wipe undamaged items with a disposable wipe; enclose damaged items in an impervious container labeled "Hazardous." Deactivate, decontaminate, clean the C-PEC; discard mat and disposables as hazardous waste.

⊗ Damaged HD packages are SPILLS — report to the designated person and follow spill SOPs.

⚠ A damaged HD shipment isn't a delivery problem — it's a contamination event.

Labeling, Packaging, Transport, and Disposal

Labeling

- HDs requiring special handling must be clearly labeled at all times during transport
- Labeling processes must NOT introduce contamination into non-HD areas

Packaging

- Must maintain physical integrity, stability, and sterility (if needed)
- Must protect both the HD and the healthcare worker who transports it
- SOPs define appropriate shipping containers, insulating materials, and mode of transport

Transport

- Comply with all applicable federal, state, and local regulations
- Use containers that minimize risk of breakage or leakage
- Pneumatic tubes must **NOT** be used to transport liquid HDs or any antineoplastic HDs
- When shipping outside the entity, consult the SDS Transport section; labels must include storage, disposal, and HD-category information

Disposal

- All custodial waste-removal personnel in HD areas must be trained
- Dispose of all HD waste — including unused HDs and trace-contaminated PPE/materials — per all applicable regulations

 Pneumatic tubes are a categorical "no" for liquid and antineoplastic HDs.

Dispensing, Compounding, and Administering

Dispensing Final Dosage Forms

- HDs requiring no further manipulation beyond counting or repackaging may be dispensed without additional containment, unless visible HD dust or leakage is present, or unless the manufacturer requires otherwise
- Use dedicated, decontaminated equipment
- Tablet/capsule forms of antineoplastic HDs must **NOT** be placed in automated counting/packaging machines

Compounding

- Must comply with <795> and <797> as applicable
- Use a plastic-backed prep mat in the C-PEC; change after spills and at end of daily compounding
- Dedicated mortars, pestles, spatulas, and other compounding equipment
- Handle APIs and powdered HDs in a C-PEC, especially during particle-generating activity

Administering

- Use protective devices and techniques (needleless systems, closed systems, plastic-pouch crushing)
- CSTDs **MUST** be used for administration of antineoplastic HDs when the dosage form allows
- Wear appropriate PPE; remove and dispose at the administration site in a trace-contaminated waste container
- Avoid manipulating HDs (crushing/opening) when possible — prefer liquid formulations

 Across dispensing, compounding, and administering — dedicated equipment, containment-first technique, and trace-waste discipline.

The Four Cleaning Steps

Sterile compounding adds a fourth step — disinfection — after the first three. These steps are sequential and non-substitutable.

Table 5 — Deactivation, Decontamination, Cleaning, and Disinfection

Step	Purpose	Example Agents
1. Deactivation	Render the compound inert or inactive	Per HD labeling; EPA-registered oxidizers (peroxide formulations, sodium hypochlorite)
2. Decontamination	Remove HD residue	Validated agents — alcohol, water, peroxide, sodium hypochlorite
3. Cleaning	Remove organic and inorganic material	Germicidal detergent
4. Disinfection (sterile only)	Destroy microorganisms	EPA-registered disinfectant and/or sterile alcohol

Important Rules

- Apply agents via **wipes** — NOT spray bottles — to avoid spreading residue
- If sodium hypochlorite is used, neutralize with sodium thiosulfate or follow with sterile alcohol/water/germicidal detergent to prevent corrosion of stainless steel
- C-PEC must be decontaminated at least **DAILY** when used; between different HDs; before and after certification; after voluntary interruption; if the ventilation tool is moved
- Area under the C-PEC work tray: deactivated, decontaminated, and cleaned at least **MONTHLY** (consider respiratory protection — containment airflows are compromised)
- PPE: two pairs of chemotherapy gloves + impermeable disposable gown; add eye/face protection if splashing is likely; respiratory protection if warranted

 The four steps are sequential and non-substitutable — disinfection without prior cleaning is meaningless.

Spill Control

Spill Kit Requirements

- Readily available in **all areas** where HDs are routinely handled
- A kit AND a respirator must be available in non-routine areas where HDs are prepared or administered
- All spill materials disposed of as **hazardous waste**

Personnel

- Only qualified, trained personnel with appropriate PPE may clean spills
- Qualified personnel must be available at all times while HDs are being handled
- Personnel with potential exposure or direct skin/eye contact require **IMMEDIATE evaluation**
- Non-employees exposed to a spill should follow entity policy (typically emergency services, incident report, exposure form)

SOP Must Address

- Size and scope of spill, who is responsible, required PPE
- Spill-management steps (decontamination, deactivation, cleaning)
- Location and capacity of spill kits
- Use of full-facepiece chemical-cartridge respirators when spill exceeds kit capacity or vapor/gas exposure is suspected

⊗ Every spill must be documented — circumstances and management.

⚠ A spill is an exposure event — document the incident, evaluate the people, and dispose of everything as hazardous waste.

Documentation, SOPs, and Medical Surveillance

SOPs and Documentation (§17)

- SOPs maintained for **ALL** HD handling situations
- Reviewed by the designated person at least every **12 months**; review documented
- SOPs must cover: hazard communication, occupational safety, area designation, receipt, storage, compounding, engineering controls, PPE, deactivation/decontamination/cleaning/disinfection, dispensing, transport, administration, environmental monitoring, disposal, spill control, medical surveillance
- Training documented per OSHA standards (29 CFR 1910.120)

Medical Surveillance (§18)

- Enroll personnel who handle HDs as a regular part of their job
- **Baseline (pre-placement)**: medical and reproductive history, work history, physical exam, baseline labs (e.g., CBC)
- **Periodic surveillance**: updated history, physical, labs as appropriate
- Exit examination at end of employment
- Records maintained per OSHA's access-to-records rule (29 CFR 1910.1020)
- **Follow-up plan** for any health change or acute exposure: post-exposure exam, review of controls, PPE compliance review, alternative duty if needed

☐ Medical surveillance is BOTH primary (baseline) and secondary (early detection) prevention.

⚠ Surveillance is the feedback loop — it tells you whether all your other controls are actually working.

Key Takeaways

1

Three Protected Populations

<800> protects patients, workers, and the environment — every requirement maps to one of them.

2

The NIOSH List Defines HDs

The entity's list — reviewed **annually** — defines what YOU handle.

3

Full Containment vs. AoR

APIs and manipulated antineoplastics require full containment; everything else may qualify for an Assessment of Risk.

4

Layered Engineering Controls

C-PEC inside C-SEC, with CSTDs as supplemental — CSTDs are **required** for administration of antineoplastics.

5

PPE Standards

ASTM D6978 gloves doubled, changed every **30 minutes**; permeation-tested gowns changed every **2–3 hours**.

6

Two Sterile HD Paths

ISO 7 buffer + ante (full **<797>** BUDs) or C-SCA (shorter BUDs).

7

Wipe Sampling

Sample at baseline and at least every **6 months**; document and remediate any measurable contamination.

8

Spills & Surveillance

Spills are exposure events; medical surveillance is the entity's safety check.

⚠ Containment, PPE, monitoring, and surveillance — the four pillars of an <800> program.

Questions & Discussion

Open the floor for case-based questions on: AoR scenarios, engineering control selection, PPE for specific activities, spill response, and medical surveillance.

References

- USP-NF General Chapter <800>, Official 1-Jul-2020 (RB 1-Jul-2020)
- NIOSH List of Antineoplastic and Other Hazardous Drugs in Healthcare Settings (current edition)
- OSHA 29 CFR 1910.1200 (Hazard Communication)
- OSHA 29 CFR 1910.134 (Respiratory Protection)
- OSHA 29 CFR 1910.120 (HAZWOPER)
- OSHA 29 CFR 1910.1020 (Access to Records)
- Companion USP chapters: <795> [Nonsterile Compounding](#) · <797> [Sterile Compounding](#)

Acronym Reference

ACPH · API · ASTM · BSC · BUD · CACI · CAI · C-PEC · C-SCA · C-SEC · CSP · CSTD · CVE · GHS · HD · HEPA · LAFW · NIOSH · OSHA · PAPR · PPE · SDS · SOP · ULPA · UPS

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